

Shayan Shakeri

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Toronto, Canada · Canadian citizen · TN-eligible (no sponsorship required)

PROFESSIONAL SUMMARY

AI Research Engineer focused on the science of measurement: evaluations and graders that resist gaming. At BMO, built a deterministic agent eval pipeline that surfaced a systematic risk-downplaying bias in a GenAI tool serving \$200B+ AUM; at Merlyn Labs, design VLM-judge graders that turn rollouts into dense, hard-to-game RL rewards and open-sourced a flow-matching VLA integration for RLinf building RL environments on BEHAVIOR-1K in OmniGibson.

CORE COMPETENCIES

Eval's & Graders

RL Environments

Reward Design (Hard-to-Game)

Post-training / RLHF & RLAIF

Measurement Science (Reliability/Variance)

LLM & Tool-Using Agents

WORK EXPERIENCE

BMO AI Centre of Excellence

Sep 2024 - Present

AI Research Engineer | Toronto, Canada

- Built a **deterministic agent eval pipeline** over hundreds of synthesized inputs to detect misaligned outputs at scale.
- Used it to uncover a systematic bias to downplay investment risk in a GenAI tool serving **\$200B+ AUM**.
- Building a graph-based **agentic system** answering complex multi-hop queries across bank client data (tool-using agents).

Merlyn Labs

Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Design **VLM-judge graders** scoring rollouts into dense, context-dependent RL rewards that are difficult to game.
- Open-sourced a **flow-matching VLA integration for RLinf**, building RL environments on the BEHAVIOR-1K suite in OmniGibson.
- Post-training result:** conservative finetuning doubled position-swap success (21% to 42%), evidence of spatial generalization over memorization.

Epineuron Technologies

May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Authored validation protocols (IEC 60601-1, ISO 13485) for PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials.

PROJECTS

Stanford BEHAVIOR-1K Challenge

8th Place | Standard Track

Identified proprioceptive collapse as a critical VLA failure mode: 60% proprioception masking improved task success by up to 48%. Trained on 10,000+ demonstrations across 22 tasks; published a technical report and LessWrong failure-mode analysis.

Measurement science | Failure-mode discovery | RL | Imitation learning

EDUCATION

M.Eng in AI & Robotics | University of Toronto

Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics (Co-op) | McMaster University

Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Eval's & Alignment: Deterministic Eval Pipelines, VLM Judges / Graders, Alignment Evaluation, Reward Design, Synthetic Data, RL Environments (OmniGibson), Post-training (RL / Fine-tuning)

ML & AI: Python, C/C++, PyTorch, JAX, RL, RLHF/RLAIF, Imitation Learning, LLM Fine-tuning, Agents

Robotics: VLA Models, Flow Matching, MuJoCo, Sim2Real, ROS2